



# IndianOil

## JOURNEY RISK MANAGEMENT (JRM) STUDY

### Salem Terminal to DURAI NAL FUEL

#### Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

#### Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

#### Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

## Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

## Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

## Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

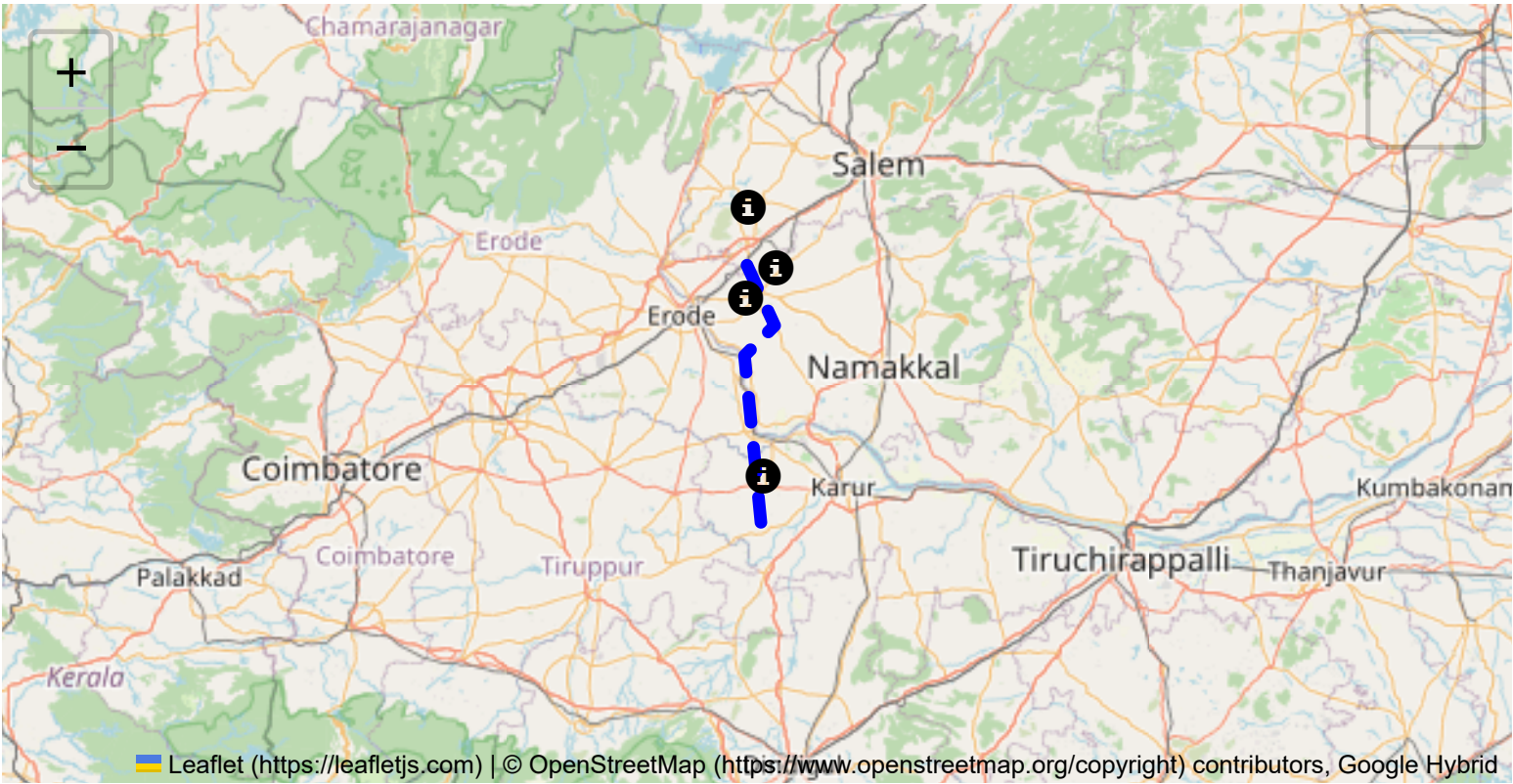
## Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

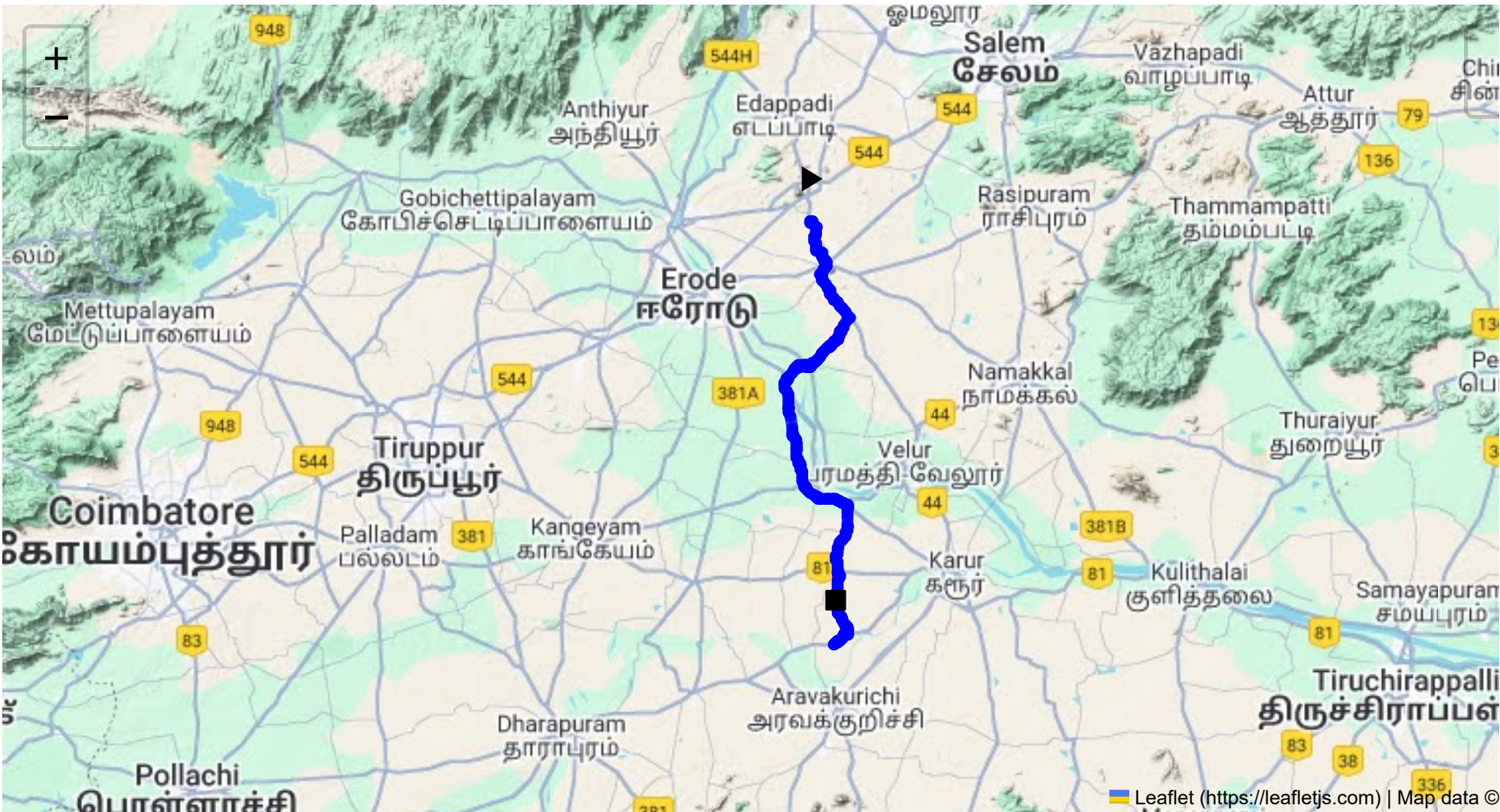
## Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.





**Route Summary:**  
**Total Distance: 80.64 km**  
**Estimated Duration: 1.9 hours**  
**Adjusted Duration (Heavy Vehicle): 2.4 hours**  
**Start: (11.4381, 77.8734)**  
**End: (10.860567, 77.906068)**



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route from CVQF+23W, Sangagiri, to VW64+FMR, Karur, encompasses key areas: SH 86 in Sullipalayam and Solasiramani - Pasur. This route is approximately 80.64 kilometers and is estimated to take about 1.91 hours for trucks carrying hazardous materials. The journey involves navigating through rural and semi-urban areas typical of Tamil Nadu roadways, along predominantly state highways and local roads.



## 2. Typical Weather Conditions and Potential Weather-Related Hazards

Tamil Nadu generally experiences a tropical climate. The region can expect hot and dry conditions from March to June, and heavy rainfall from October to December due to the northeast monsoon. Weather-related hazards include:

- **Monsoon Flooding:** Roads may become waterlogged, obscuring visibility and affecting road conditions.
- **Summer Heat:** Potential for overheating vehicles. Check coolant levels frequently.

## 3. Traffic Patterns and Congestion

There is moderate traffic on SH 86 with peak congestion during:

- **Mornings (8:00 AM - 10:00 AM):** School and commuter traffic.
- **Evenings (5:00 PM - 7:00 PM):** Return commuters and local market activities.

Sullipalayam and Solasiramani areas may experience localized congestion due to narrow roads and roadside markets.

## 4. Assessment of Road Quality and Infrastructure

The main route via SH 86 is reasonably well-maintained, but expect:

- **Potholes** and uneven surfaces, especially after monsoon.
- **Narrow Sections:** Some segments in rural areas can be narrow, requiring caution.
- **Limited Signage:** Be attentive to road markers and village entry points.

## 5. Alternative Routes for Emergencies

In case of emergencies, such as roadblocks or accidents:

- Route 1: Divert to NH 44, a major national highway, though it is longer.
- Route 2: Use SH 21 connecting Erode and Karur as an alternative.

## 6. Local Regulations on Hazardous Material Transport

- Strict adherence to vehicle permits for transporting hazardous materials.
- Compliance with local speed regulations.
- Restrictions on transporting hazardous materials during certain hours in densely populated areas.

## 7. Historical Incidents

There have been sporadic incidents involving heavy vehicles, typically focusing on:

- **Overtaken Trucks:** Due to sharp bends and narrow roads.

- **Spillages:** Lack of containment measures during accidents.

## 8. Environmental Considerations and Sensitive Areas

Focus on minimizing the impact on:

- **Agricultural Fields:** Risk of contamination from spills.
- **Water Bodies:** Bridges and crossings should be navigated with care to prevent pollution.

## 9. Communication Coverage

Relatively good mobile coverage along SH 86 with potential dead zones around:

- **Remote areas near Solasiramani.**
- **Isolated stretches with dense vegetation.**

## 10. Estimated Emergency Response Times

Emergency services are generally available but could be delayed in rural segments:

- **Urban Areas:** 15-30 minutes.
- **Rural Areas:** 45-60 minutes, depending on proximity to the nearest town.

## 12. Overall Summary of Risk Assessment

The route is generally manageable for transporting hazardous materials but poses risks such as:

- **Weather Vulnerability:** During monsoons.
- **Road Quality:** Impacting vehicle handling.
- **Traffic Management:** Due to narrow roads in village segments.
- **Limited Emergency Services Accessibility:** Impacts response time.
- **Compliance with Regional Regulations:** Necessary to avoid penalties and ensure safety.

Drivers should proceed with caution, particularly in bad weather and during village passages. Regular vehicle checks, preparation for unexpected weather, and adherence to route regulations will minimize many potential hazards.

### Risk Assessment - Turns

|   | Risk Type  | Risk Level | Coordinates        | Speed Limit |
|---|------------|------------|--------------------|-------------|
| 0 | Roundabout | High       | 11.38395, 77.89480 | 15 KM/Hr    |
| 1 | Turn       | High       | 11.43968, 77.87345 | 15 KM/Hr    |
| 2 | Turn       | High       | 11.44019, 77.87547 | 15 KM/Hr    |

|    | Risk Type  | Risk Level | Coordinates        | Speed Limit |
|----|------------|------------|--------------------|-------------|
| 3  | Turn       | Medium     | 11.39685, 77.88861 | 30 KM/Hr    |
| 4  | Turn       | High       | 11.37862, 77.89488 | 15 KM/Hr    |
| 5  | Turn       | High       | 11.37850, 77.89453 | 15 KM/Hr    |
| 6  | Turn       | High       | 11.30859, 77.92918 | 15 KM/Hr    |
| 7  | Turn       | Medium     | 11.24192, 77.87363 | 30 KM/Hr    |
| 8  | Blind Spot | Blind Spot | 11.24001, 77.87368 | 10 KM/Hr    |
| 9  | Turn       | High       | 11.24061, 77.86899 | 15 KM/Hr    |
| 10 | Turn       | Medium     | 11.24111, 77.86893 | 30 KM/Hr    |
| 11 | Turn       | Medium     | 11.24127, 77.86879 | 30 KM/Hr    |
| 12 | Turn       | High       | 11.24157, 77.86876 | 15 KM/Hr    |
| 13 | Turn       | High       | 11.24163, 77.86852 | 15 KM/Hr    |
| 14 | Turn       | High       | 11.24242, 77.86833 | 15 KM/Hr    |
| 15 | Turn       | Medium     | 11.24248, 77.86799 | 30 KM/Hr    |
| 16 | Blind Spot | Blind Spot | 11.24384, 77.86689 | 10 KM/Hr    |
| 17 | Turn       | High       | 11.24100, 77.86026 | 15 KM/Hr    |
| 18 | Turn       | Medium     | 11.24194, 77.85910 | 30 KM/Hr    |
| 19 | Turn       | Medium     | 11.24199, 77.85864 | 30 KM/Hr    |
| 20 | Turn       | High       | 11.22377, 77.83980 | 15 KM/Hr    |
| 21 | Turn       | Medium     | 11.21072, 77.83922 | 30 KM/Hr    |
| 22 | Turn       | Medium     | 11.12703, 77.85401 | 30 KM/Hr    |
| 23 | Turn       | Medium     | 11.05046, 77.92522 | 30 KM/Hr    |
| 24 | Turn       | Medium     | 10.95908, 77.91206 | 30 KM/Hr    |
| 25 | Turn       | High       | 10.95890, 77.91189 | 15 KM/Hr    |
| 26 | Turn       | High       | 10.95883, 77.91107 | 15 KM/Hr    |
| 27 | Turn       | Medium     | 10.95606, 77.91351 | 30 KM/Hr    |
| 28 | Turn       | Medium     | 10.95170, 77.91380 | 30 KM/Hr    |
| 29 | Turn       | Medium     | 10.91844, 77.91242 | 30 KM/Hr    |
| 30 | Turn       | Medium     | 10.91798, 77.91254 | 30 KM/Hr    |
| 31 | Turn       | High       | 10.91757, 77.91295 | 15 KM/Hr    |
| 32 | Turn       | Medium     | 10.91525, 77.91232 | 30 KM/Hr    |
| 33 | Turn       | Medium     | 10.90953, 77.91356 | 30 KM/Hr    |
| 34 | Turn       | Medium     | 10.90217, 77.91353 | 30 KM/Hr    |
| 35 | Turn       | High       | 10.89444, 77.91835 | 15 KM/Hr    |
| 36 | Turn       | High       | 10.89425, 77.91891 | 15 KM/Hr    |

|    | Risk Type | Risk Level | Coordinates        | Speed Limit |
|----|-----------|------------|--------------------|-------------|
| 37 | Turn      | Medium     | 10.88533, 77.92242 | 30 KM/Hr    |
| 38 | Turn      | Medium     | 10.88353, 77.92517 | 30 KM/Hr    |
| 39 | Turn      | High       | 10.87500, 77.92565 | 15 KM/Hr    |

## Emergency Locations

|    | type     | name                             | coordinates            | speed_limit | risk_level |
|----|----------|----------------------------------|------------------------|-------------|------------|
| 0  | hospital | Tiruchengode, Goverment Hospital | 11.3903328, 77.8920627 | 30 km/h     | Medium     |
| 1  | hospital | SPM Medical Centre,Tiruchengode  | 11.3881331, 77.8931963 | 30 km/h     | Medium     |
| 5  | hospital | T.C.A Hospital Tiruchengode      | 11.3791885, 77.8965774 | 30 km/h     | Medium     |
| 6  | clinic   | Kongu Nursing Home               | 11.3783065, 77.8961134 | 30 km/h     | Medium     |
| 7  | hospital | Soorya Multispecialty Hospital   | 11.3786429, 77.8931912 | 30 km/h     | Medium     |
| 8  | hospital | Tiruchengode Government Hospital | 11.37645, 77.89426     | 30 km/h     | Medium     |
| 9  | hospital | Krishna Hospital, Namakkal       | 11.3754811, 77.8931817 | 30 km/h     | Medium     |
| 10 | hospital | Tirukumaran Hospitals            | 11.3782118, 77.8914933 | 30 km/h     | Medium     |
| 13 | police   | கரட்டுப்பாளையம் காவல் நிலையம்    | 11.357586, 77.8922539  | 30 km/h     | Medium     |
| 14 | clinic   | Jai Sri Amman Clinic             | 11.1102489, 77.8579522 | 30 km/h     | Medium     |

## Crowded Spots

|   | type   | name                         | coordinates            | speed_limit | risk_level |
|---|--------|------------------------------|------------------------|-------------|------------|
| 2 | school | அரசு ஆண்கள் மேல்நிலைப் பள்ளி | 11.3850439, 77.8948279 | 30 km/h     | Medium     |



|    | type        | name                                | coordinates               | speed_limit | risk_level |
|----|-------------|-------------------------------------|---------------------------|-------------|------------|
| 3  | school      | அரசு பெண்கள் மேல்நிலைப் பள்ளி       | 11.3844247,<br>77.8949053 | 30 km/h     | Medium     |
| 4  | marketplace | திருச்செங்கோடு தினசரி காய்கறி சந்தை | 11.3833608,<br>77.8970145 | 30 km/h     | Medium     |
| 11 | school      | KSR Educational institution         | 11.3772013,<br>77.8908807 | 30 km/h     | Medium     |
| 12 | school      | MDV School                          | 11.3719843,<br>77.8915244 | 30 km/h     | Medium     |

Route Photos of Risky Spots



Risk Type: Roundabout  
Risk Level: High  
Speed Limit: 15 KM/Hr  
Coordinates: 11.38395, 77.89480



Risk Type: Turn  
Risk Level: High



Speed Limit: 15 KM/Hr  
Coordinates: 11.44019, 77.87547



Risk Type: Turn  
Risk Level: Medium  
Speed Limit: 30 KM/Hr  
Coordinates: 11.39685, 77.88861



Risk Type: Turn  
Risk Level: High  
Speed Limit: 15 KM/Hr  
Coordinates: 11.37862, 77.89488





Risk Type: Turn  
Risk Level: High  
Speed Limit: 15 KM/Hr  
Coordinates: 11.37850, 77.89453



Risk Type: Turn  
Risk Level: High  
Speed Limit: 15 KM/Hr  
Coordinates: 11.30859, 77.92918





**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.24192, 77.87363



**Risk Type:** Blind Spot

**Risk Level:** Blind Spot

**Speed Limit:** 10 KM/Hr

**Coordinates:** 11.24001, 77.87368



**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.24061, 77.86899





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.24111, 77.86893



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.24127, 77.86879





**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.24157, 77.86876



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.24163, 77.86852



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.24242, 77.86833





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.24248, 77.86799



**Risk Type:** Blind Spot  
**Risk Level:** Blind Spot  
**Speed Limit:** 10 KM/Hr  
**Coordinates:** 11.24384, 77.86689





**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.24100, 77.86026



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.24194, 77.85910



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.24199, 77.85864





**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.22377, 77.83980



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.21072, 77.83922





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.12703, 77.85401



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.05046, 77.92522



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 10.95908, 77.91206





**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 10.95890, 77.91189



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 10.95883, 77.91107





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 10.95606, 77.91351



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 10.95170, 77.91380



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 10.91844, 77.91242





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 10.91798, 77.91254



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 10.91757, 77.91295





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 10.91525, 77.91232



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 10.90953, 77.91356



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 10.90217, 77.91353





**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 10.89444, 77.91835



**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 10.89425, 77.91891





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 10.88533, 77.92242



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 10.88353, 77.92517



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 10.87500, 77.92565



